Playbook Variant Validation Guide

Repository: agentic_ai_updated_with_resource_alloc_without_hardcoding_with_human_review_updated_1

Folder: data-for-enhancement

Purpose

These five playbook variants are controlled business-policy tests for the SeeWeeS agent. Each variant changes one policy area while leaving the code untouched. The goal is to verify that special-case review, weather escalation, SLA priority, capacity planning, and penalty scoring are driven by the playbook rather than hardcoded assumptions.

Variant File	Policy Change	Expected Agent Behavior	Business Significance
playbook_variant_01_special_case_item.md	In Appendix A.3, the SPECIAL_CASE row is changed from item_id 99999 to item_id 10040 mapped to EPI-AI.	Human review should trigger for item_id 10040 when it appears in planning-window shipments. The trigger should not depend on item_id 99999.	Validates that high-risk clinical/business review controls are policy-driven from the playbook, not hardcoded to one item.
playbook_variant_02_weather_escalation_score2.md	In the travel buffer table, risk score 2 is changed from +25% buffer to +25% buffer + escalation.	The severe-weather human review threshold should become score 2 because the policy parser identifies the first travel-buffer row containing escalation.	Validates that weather escalation governance can be tightened by policy update without code changes.
playbook_variant_03_corridor_sla_swap.md	In the corridor catalog, C1_I95_NJ_BOS is changed to Tier 2 and C2_NJ_PHL is changed to Tier 1.	When scarce resources exist, allocation priority and penalty interpretation should shift toward C2_NJ_PHL as the Tier 1 corridor.	Validates that corridor priority and SLA strategy are controlled by the playbook, supporting executive policy changes.
playbook_variant_04_tighter_truck_capacity.md	Truck capacity is reduced from 10 to 6 volume units, and packing buffer is increased from +10% to +20%.	Required truck counts should increase for affected corridor/day demand, potentially creating more shortfalls and penalties.	Validates that capacity planning responds to operational constraints such as smaller vehicles, packing inefficiency, or stricter load rules.
playbook_variant_05_penalty_weights.md	Tier 2 SLA violation penalty is increased from 40 to 90 points; cold-chain violation penalty is increased from +80 to +150 points.	Penalty totals and report rationale should reflect the new business cost model when units are undelivered or cold-chain constrained.	Validates that executive tradeoff scoring is configurable and can reflect changing customer, financial, or clinical risk priorities.

How To Use These Variants

- Start from the baseline playbook and run the normal flow to capture baseline behavior.

- Point src/main.py to one variant file at a time by changing the pdf_path value.

- Run the agent and record the human-review prompts, resource allocation, penalty scores, and report explanation.

- Compare the actual behavior to the expected behavior in this guide.

- Use the results as validation evidence that the agent follows playbook policy changes without Python code edits.

Suggested Validation Log Columns

Variant	Expected Change	Observed Output	Pass/Fail	Business Interpretation
Variant 1	Human review should trigger for item_id 10040 when it appears in planning-window shipments. The trigger should not depend on item_id 99999.	Fill after run	Fill after run	Validates that high-risk clinical/business review controls are policy-driven from the playbook, not hardcoded to one item.
Variant 2	The severe-weather human review threshold should become score 2 because the policy parser identifies the first travel-buffer row containing escalation.	Fill after run	Fill after run	Validates that weather escalation governance can be tightened by policy update without code changes.
Variant 3	When scarce resources exist, allocation priority and penalty interpretation should shift toward C2_NJ_PHL as the Tier 1 corridor.	Fill after run	Fill after run	Validates that corridor priority and SLA strategy are controlled by the playbook, supporting executive policy changes.
Variant 4	Required truck counts should increase for affected corridor/day demand, potentially creating more shortfalls and penalties.	Fill after run	Fill after run	Validates that capacity planning responds to operational constraints such as smaller vehicles, packing inefficiency, or stricter load rules.
Variant 5	Penalty totals and report rationale should reflect the new business cost model when units are undelivered or cold-chain constrained.	Fill after run	Fill after run	Validates that executive tradeoff scoring is configurable and can reflect changing customer, financial, or clinical risk priorities.